

Definite integrals



1

a 33

b -12

c 46

d -11

e -9

f 192

g 24

h 84

i 15

j 30

k 490

l 48

m 64

n $-\frac{729}{4}$

o $\frac{38}{3}$

p $\frac{59}{3}$

q 1010

r 78

s $\frac{83}{24}$

Properties of definite integrals

2

a 0

b 0

c 0

d 0

3

a 9

b 45

c 45

d Function multiplied by a constant: $\int_a^b k f(x) dx = k \int_a^b f(x) dx$

4 10

5 13

6

a 144

b -144

c Switch end points: $\int_a^b f(x) dx = -\int_b^a f(x) dx$

7

a -3

b 9

c 13

8

a 12

b -12

c 32

9

a 3

b -5

c 12



Signed areas

10

a 2

b 6

c 1

d 9 units^2

e $\int_0^6 f(x) dx$

11

a 24

b 10

c 4

d 38 units^2

e $\int_0^8 f(x) dx$

12

a -15

b -6

c -21

d 21 units^2

e $-\int_0^7 f(x) dx$

13

a 8

b -18π

c $8 - 18\pi$

d $(8 + 18\pi) \text{ units}^2$

e $\int_0^4 f(x) dx - \int_4^{16} f(x) dx$

14

a Below

b -35

c 35 units^2

15

a $x = 4$

b $x > 4$

c -36

d 36 units^2

16

a $x = 3, x = 9$

b $3 \leq x \leq 9$

c -36

d 36 units^2

17

a $x = -2, x = -8$

b $-8 \leq x \leq -2$

c 36

d 36 units^2



Area under curve

18 21 units^2

19 18 units^2

20 9 units^2

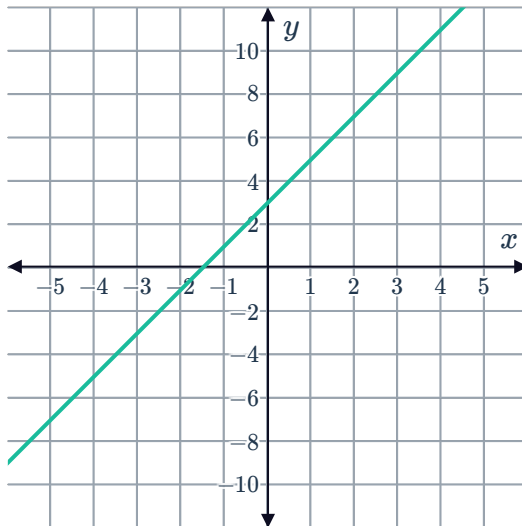
21 $\frac{23}{3} \text{ units}^2$

22 $\frac{71}{6} \text{ units}^2$

23

a

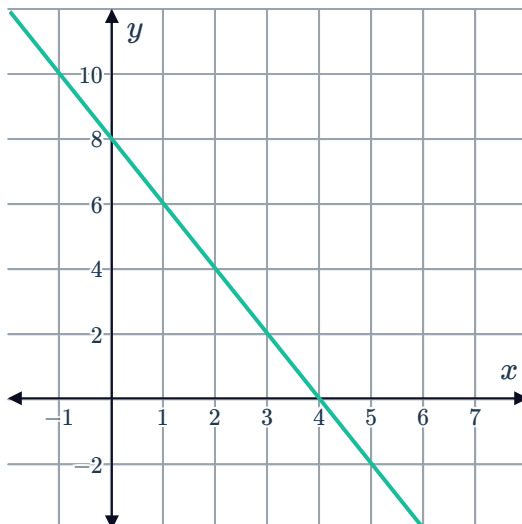
i



ii 14 units^2

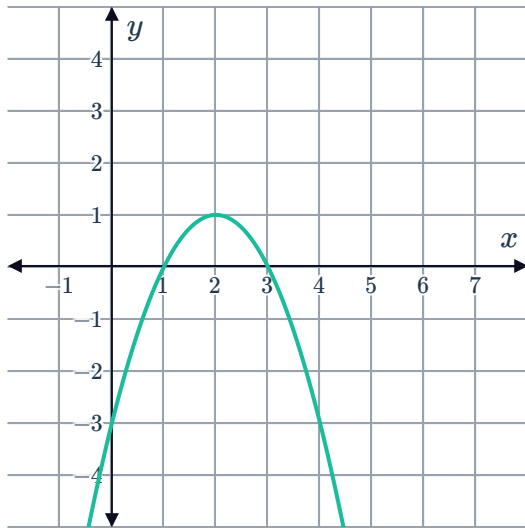
b

i



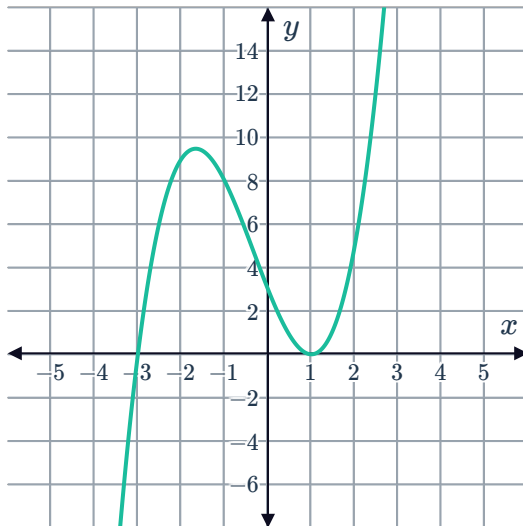
ii 8 units^2

c i



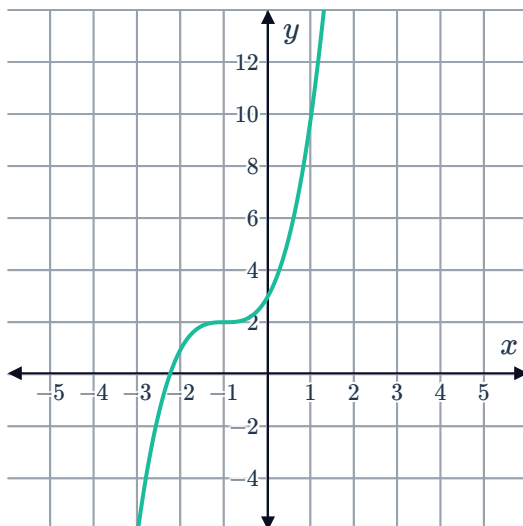
ii $\frac{4}{3}$ units²

24 a



b $\frac{33}{4}$ units²

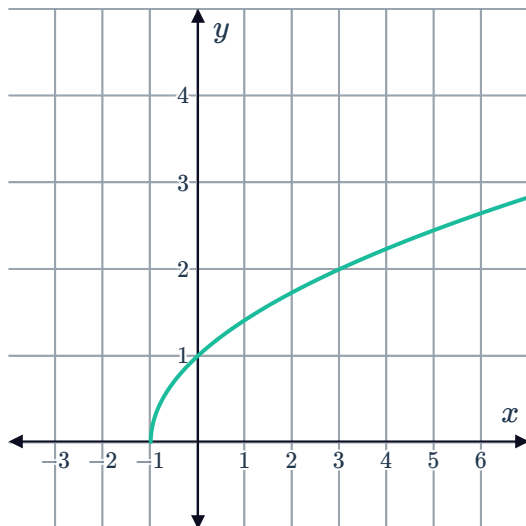
25 a



b $\frac{39}{4}$ units²

26

a



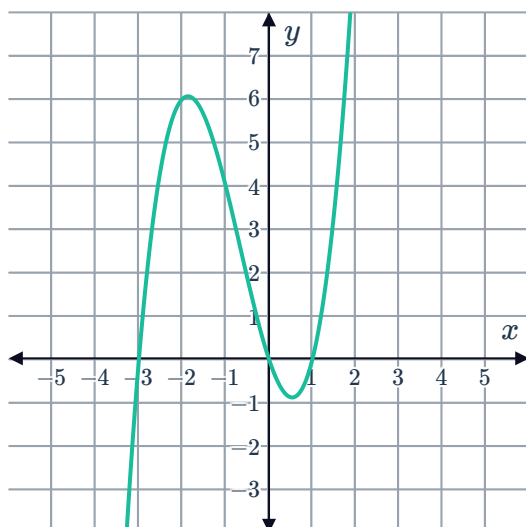
b

$$\frac{16}{3} \text{ units}^2$$

27

a

i

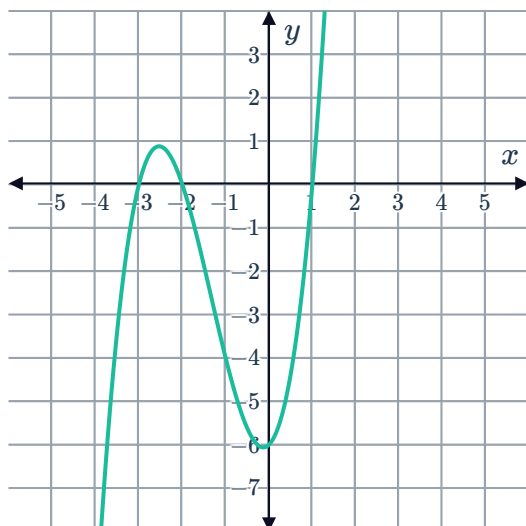


ii

$$\frac{71}{6} \text{ units}^2$$

b

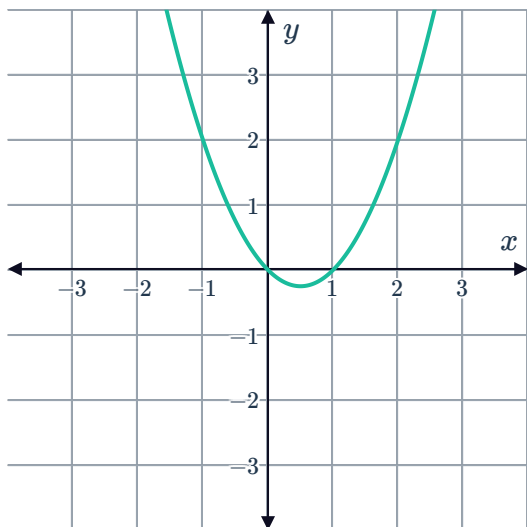
i



ii

$$\frac{71}{6} \text{ units}^2$$

c i



ii $\frac{1}{6} \text{ unit}^2$

28

a -9

b -5

c 9 units²

d 13 units²

Area as an absolute value

29

a 8

b -10

c -2

d 18 units²

e $\int_0^3 f(x) dx - \int_3^8 f(x) dx$

f $\int_0^8 |f(x)| dx$

30

a 10

b 14 units²

c $\int_2^8 f(x) dx - \int_0^2 f(x) dx$

d $\int_0^8 |f(x)| dx$

31

a -1

b 1

c 13

d 13 units²

32

a -2

b 8

c -4

d -14

e 14

f 14 units²

33

a 1

b 6

c -9

d 10 units²

e 10

f $\frac{109}{3}$

34 a $-A + B$ b $-3A - B$  c $|-A + B|$ d $A + B$
e $-A + B - 12$

35 a $B - C$ b $A - B + C$
c $(A + B + C) \text{ units}^2$ d $A + B + C$
e $35 + A - B + C$

36 a -3 b 11 c 11 units^2 d $-\frac{163}{4}$

37 a 8 b 13 units^2 c -174

38 a $|-A + B - C|$ b $(|A| + |B| + |C|) \text{ units}^2$
c $-C + A$ d $-2A + 2B - \frac{C}{2}$
e $\frac{91}{2} + A - B + C$